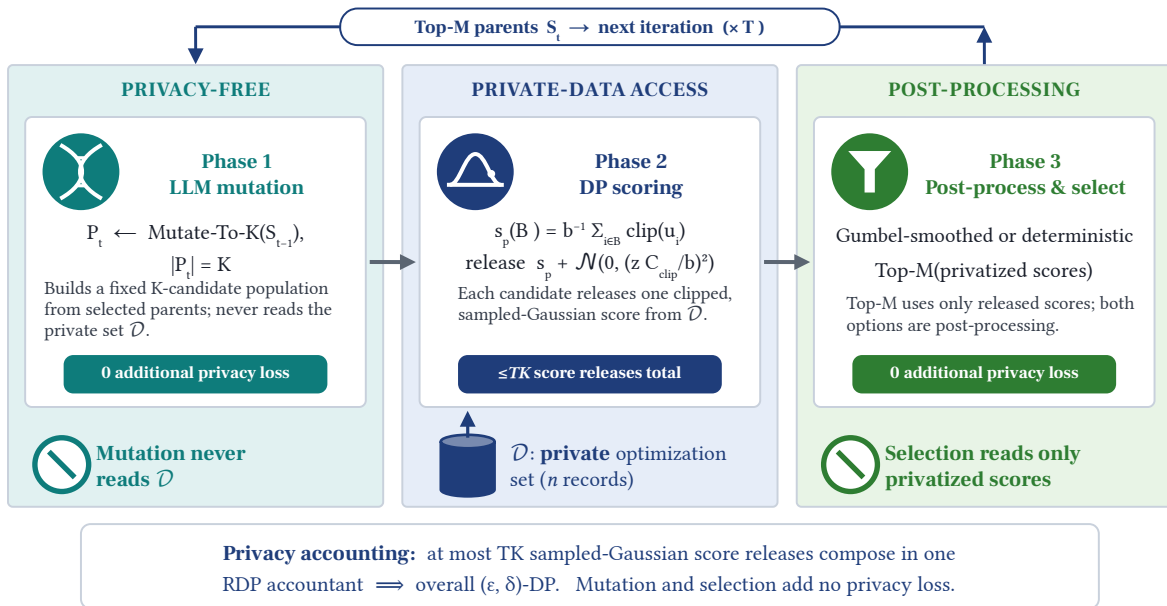


(a) One DP-ES iteration: privacy-free exploration, private scoring



(b) DP-ES vs. DP-OPT: mechanism and observed stability

Aspect	DP-ES (ours)	DP-OPT
Search unit	Population of complete prompts	Greedy token-by-token construction
Private-data access	Sampled utility scoring only	Private histogram queries during construction
Selection / recovery	Top-M on privatized scores; alternatives persist	Private token releases; no token-level backtracking
GSM8K (30 runs)	88.1 $\pm$ 3.2%	49.5 $\pm$ 28.5%